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The Editor-in-Chief
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Subject: Submission of the manuscript "Diffusion Language Models and Unified Multimodal Generation: A Critical Systematic Review of Discrete and Continuous Approaches"

Dear Editor-in-Chief,

On behalf of all authors, I am pleased to submit this manuscript as a long survey paper to ACM Computing Surveys.

Diffusion language models, and unified models that also generate images, are a fast-growing alternative to autoregressive generation, yet their reported results depend heavily on how they are decoded, and papers rarely state the decoding rule behind a headline number. Building on a systematic search of 79,963 records and 615 included studies, the survey makes the following technical contributions.

Contributions and key findings:

- **A design space that organises the field.** Every generator is described by five design choices, from the corruption channel to the commitment and revision rules; the design space places all 47 models tested, 36 fully and 11 with a stated extension.
- **A reporting gap with direct consequences.** A census of a random sample of papers finds that 147 of 153 report their headline result with a decoding rule other than plain sampling or do not state the rule, so many cross-paper comparisons are not like for like.
- **Evidence synthesis with explicit verdicts.** Comparisons with autoregressive models are split and only 1 of 25 is compute-matched, training gains are rarely isolated, and more inference compute helps most curves in 9 of 15 sweeping studies; each verdict is tested for robustness.
- **Practical guidance and open problems.** An error budget classifies methods by the error they change; a decision framework, a reporting checklist and open problems with the measurements that would settle them complete the review.

We believe the work suits the broad readership of ACM Computing Surveys across natural language processing, generative modelling and multimodal learning, serving both researchers entering the area and practitioners choosing a model or decoder. The main paper is 34 pages including references in the ACM Surveys format, with a marked electronic supplement. The protocol, records, analysis scripts and manuscript sources are archived at https://doi.org/10.5281/zenodo.23524470, so every reported number can be reproduced.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere; an earlier version of its sources is part of the data archive above, not of any journal, proceedings or preprint server. All authors have approved the manuscript and agree with its submission. The review analyses published literature only; no human participants were involved. The use of AI tools is described in the manuscript's Declaration on the use of AI. No funding was received; the article processing charge is covered by Vellore Institute of Technology, Vellore, India, and the authors declare no conflicts of interest.

Thank you for considering our manuscript. We look forward to your response.

Yours sincerely,

Prof. Subramaniyaswamy Vairavasundaram (Corresponding Author)
on behalf of Shlok Goenka and Balaji G N